

Comprehensive Fitness Data Analysis and Heart Disease Prediction

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1. INTRODUCTION:

Health and fitness data collection and analysis are being revolutionized by fitness apps such as Endomondo. With user permission, these applications gather a plethora of data, such as biometrics, exercise metrics, and user-supplied background data. In this paper, we investigate in detail the Endomondo fitness tracking data collecting procedure. Time-series sensor data from several user workout sessions, including heart rate, GPS tracks, and accelerometer data, are included in the raw data. Self-reported demographic and health information is also included.

Our analysis focuses on usage trends, adoption across age groups and locations, activity preferences, heart rate zones, and correlations between activity levels and health outcomes. We use a variety of techniques, including statistical analysis, correlation analysis, and visualizations. This research provides useful information for app developers aiming to improve user features and forms the basis for more focused modeling efforts. Specifically, our initiative focuses on analyzing heart rate data to find abnormalities that could point to underlying heart problems. To extract time and frequency domain information from heart rate time series, we use signal processing and associated techniques. This allows us to train classification models with random forests and naive bayes machine approaches to identify cardiac anomalies. Our study ultimately demonstrates how machine learning can extract important health information about heart health from fitness tracker data, providing a methodological basis for applications utilizing commercial fitness data that use preventive health screening.

KEYWORDS:

Fitness application, statistical analysis, correlation analysis data analysis, data collection, random forests, naive bayes.

2. RELATED WORK:

The prediction of heart disease has been approached in a variety of ways using machine learning algorithms. A hidden Naïve Bayes classifier is the foundation of Jabbar and Samreen's [1] Heart Disease Prediction System, which shows how well this method predicts cardiac abnormalities.

The performance of the K-Nearest Neighbors and Support Vector Machine algorithms was compared by Saikumar et al. [2], offering insights into the suitability of various models for heart disease prediction. The use of Random Forest and Long Short-Term Memory (LSTM) networks was investigated by Liu et al. [3], who emphasized the potential of deep learning architectures and ensemble methods in improving prediction accuracy.

S. G. A et al.'s [4] investigation into the prediction of heart disease using KNN and RTC algorithms advanced knowledge of the functions of various machine learning approaches. The significance of model integration was highlighted by Naidu et al.'s [5] hybridized model for heart disease prediction, which combined several machine learning algorithms. A thorough analysis of the effectiveness of various machine learning techniques in the prediction of heart disease was carried out by Sarveshvar et al. [6], who made significant contributions to our understanding of the relative merits of these approaches.

3. DATA AND METHODOLOGY:

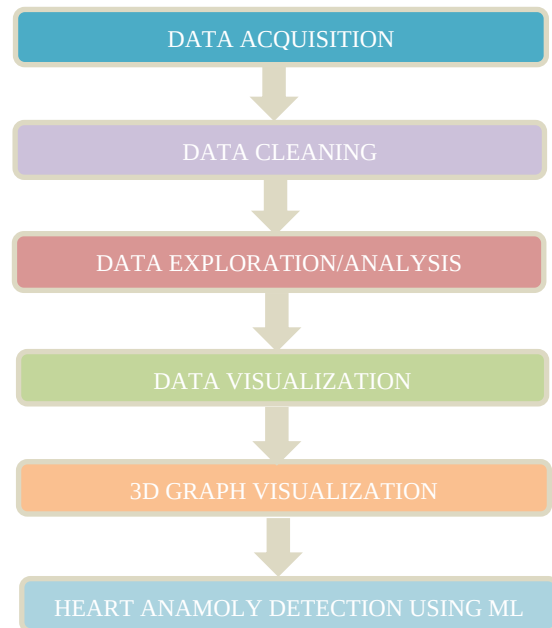


Fig 3.1 Block diagram

We carefully followed a six-step procedure for heart rate anomaly identification in this study. The first step is Data Acquisition, which entails gathering extensive heart rate data from various sources. Data dependability is then ensured by removing noise, outliers, and inconsistencies during the Data Cleaning step. Trends and correlations in the dataset are found through data exploration and analysis. Charts and plots used in data visualization successfully communicate ideas. Understanding data is deepened using 3D Graph Visualization. Finally, Heart Rate Anomaly Detection uses machine learning to spot abnormalities, improving patient care and healthcare monitoring.

3.1 Data Acquisition:

The FitRec dataset consists of user sport records from the mobile application Endomondo, which allows users to monitor their health and fitness metrics. Sequential sensor data, such as heart rate, speed, and GPS coordinates, are included, along with extra information about the user's gender, sport, and weather. Dataset is 6.57GB Json file. This dataset, which includes 253,020 workouts from 1,104 users, was gathered exclusively for scholarly research. There are two versions of it: one is raw, and the other is filtered. The filtered version is used for things like short-term heart rate prediction and workout route prediction.

```

userId: 10921915
gender: male
sport: bike
id: 396826535
longitude: [24.6497704886679, 24.65014273309767, 24.650910682976246, 24.65066865069747, 24.649145286530256, ...]
latitude: [60.173348765828265, 60.173239801079035, 60.17298021353781, 60.172477969899774, 60.17186114564538, ...]
altitude: [-1.8044666444624418, -1.8190453555595787, -1.8190453555595787, -1.8511185199732794, -1.871528715589271, ...]
timestamp: [1408898746, 1408898754, 1408898765, 1408898778, 1408898794, ...]
time_elapsed: [-0.1225675259145224, -0.12221890169596584, -0.12172054383967204, -0.12114183000958663, -0.12042778221853381, ...]
heart_rate: [-0.197369036801112, -5.867841781816304, -3.961864789919643, -4.173640002263717, -3.961864789919643, ...]
derived_speed: [-7.0829444309064396, -2.8861928357804815, -0.3976286593020398, -0.7571073884764162, 2.6415189187026646, ...]
distances: [-4.372303649217081, -2.374952819539426, -0.07926348591212737, 0.4284751220389611, 4.710835408111755, ...]
tar_heart_rate: [100, 111, 120, 119, 120, ...]
tar_derived_speed: [0, 10, 75137615573548, 16.886294372816662, 15.902596545763366, 24.446443396153843, ...]

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Fig 3.2 Example of one workout sample from fitrec dataset

3.2 Data Cleaning:

When analyzing a new dataset, one of the first stages is to look for missing values. Missing values are entries that are empty or undefined rather than containing a data point. We scan for None or null values in text and category columns. These denote an empty string type entry. We search for zeroes or NaN (Not a Number) values in numerical columns. These denote an empty numeric item. Finally, for columns that are arrays or lists of multiple values, we check to see if the arrays include any zeroes or NaNs.

3.3 PySpark Lazy evaluation:

When working with DataFrames and datasets, PySpark uses lazy evaluation, which means that computations are postponed until an event that necessitates the outcome is initiated. For example, filter and map operations are not performed right away; instead, they are saved as instructions for a later time. Due to the possibility of early filtering discarding data required for later processes, this

strategy is more efficient. Concrete results are obtained only when an operation such as count() or collect() is invoked; only then does the real filtering and mapping take place. Before performing the action, PySpark gathers all previously stored activities, deferring processing until the right time. When opposed to eager execution, lazy evaluation greatly improves speed for large data applications by streamlining several steps into a single, effective procedure.

3.4 Exploratory Data Analysis:

Fitness monitoring databases like FitRec offer valuable time-series sensor data for uncovering insights into activity patterns, health markers, and lifestyle connections. To better comprehend the dataset, exploratory data analysis (EDA) uses visuals and summary statistics. EDA plays a key role in exposing trends, physiological reactions, and activity patterns in fitness monitoring data. Plotting data distributions and computing summary statistics such as mean, standard deviation, and percentiles are usually the first steps in evaluating variables like heart rate and illuminating central trends and variability.

3.5 Unstack PySpark Dataframe:

To unstack a PySpark DataFrame in the context of FitRec data, use the pivot method, which effectively unstacks or pivots the data. The unstacking technique converts data from a long format (where numerous columns are used to represent distinct features) to a wide format, making it easier to work with and analyze. In this work, we utilized unstack to prepare the data for visualization and generate a stacked bar chart displaying the gender distribution of activity participation in each sport.

3.6 Pyspark UDF registering:

A Spark UDF (user-defined function) has the Python code wrapped around it. This enables the scalable distribution and application of Python logic across Spark datasets. The focus of this PySpark-based work is on the 'timestamp' column, which is critical for time series analysis, particularly in health-related predictions such as heart rate monitoring. This column has resulted in the following new features:

'date_time': Converts UNIX timestamps to Python datetime format, making time-based analysis easier.

'duration': The time difference between the initial and end timestamps in each session is used to calculate the overall exercise length in minutes.

'workout_start_time': The hour of the day at which a workout begins, as determined by the first timestamp record.

'interval': A collection of time gaps between successive timestamp records inside a single session that may be used to understand workout trends.

These additional characteristics improve the dataset's time-related insights, assisting in predictive analysis such as the early detection of health problems.

3.7 Convert row objects to RDD:

RDD (Resilient Distributed Dataset) in Spark is a sophisticated feature that allows data to be divided into logical divisions for parallel processing across cluster nodes, fault tolerance, and performance optimization. You can efficiently carry out distributed calculations by utilizing RDDs. To provide useful insights into data distribution and analysis, the code in PySpark attempts to extract and analyze statistics from the 'interval' column, convert them to RDD, and plot important statistics.

In this work, we have created interval statistic dataframe from tabulated statistics data by using the row objects. For plotting purposes, these row objects are transformed to Pandas DataFrames, which makes it easier to visualize interval statistics, such as quantiles, min, mean, max, and standard deviation, across several sports.

4. ML ALGORITHMS IMPLEMENTATION

Time series heart rate data from the FitRec dataset offers a valuable resource for detecting irregular heart rate patterns, which can potentially signal cardiac abnormalities. In order to discriminate between normal and abnormal heart rate patterns, each heart rate time series is exposed to machine learning models such as Random Forests, Logistic Regression, and Naive Bayes.

Preprocessing raw heart rate data includes noise reduction, missing value imputation, and extraction of pertinent heart rate features, such as heart health markers. This is the first stage in the process. The training of classification algorithms that are intended to detect anomalies is based on these derived properties. For example, Random Forests aggregate predictions from many decision trees created from training data subsets. While Naive Bayes implies conditional feature independence, Logistic Regression uses logistic functions to determine the likelihood of an anomaly occurring based on heart rate information. By using examples, these models are taught to distinguish between heart rate patterns that are normal and irregular heartbeats that may be a sign of ischemia or arrhythmias. Cross-validation and performance analysis work together to help choose the best model. Hence, FitRec users' unobserved heart rate time series may be categorized as normal or abnormal, opening the door to the creation of an early warning system for cardiac illness screening that is only dependent on fitness tracker data.

4.1. Logistic Regression

Logistic Regression is a supervised machine learning model that is commonly used for binary classification tasks. The logistic function is used to model the likelihood

of an instance belonging to a specific class. The model assumes a linear relationship between the input features and the target class's log-odds. Iterative optimization methods such as gradient descent are used to estimate parameters. The logistic function squeezes predictions between 0 and 1, interpreting them as probabilities during training. To ensure that the predictions fall between 0 and 1, the output is transformed using the logistic function (sigmoid function).

$$\text{sigmoid function} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{hypothesis function} = h_{\theta}(x) = \sigma(\theta^T x)$$

The model is trained by determining the parameter values that maximize the likelihood of the observed data which is typically accomplished using optimization algorithms such as gradient descent.

Logistic Regression is adaptable, interpretable, and computationally efficient, making it appropriate for a wide range of applications. However, features must have some degree of linearity, and outliers can have an impact on performance. Depending on the problem and requirements, model performance is measured using metrics such as accuracy, precision, recall, F1-score, and so on.

4.2. Random Forest

The Random Forest machine learning model is an ensemble approach that builds many decision trees during training and produces the mean prediction (regression) of the individual trees or the mode of the classes (classification). Using a method known as bagging (Bootstrap Aggregating), each tree is trained on a random subset of the data. The generalization of the model is enhanced, and overfitting is lessened by the randomness of the features and data.

Every decision tree in the Random Forest algorithm is built using a random subset of the training data and a random subset of features at each split. The model is made more robust and resilient to data noise when the individual trees' diversity is paired with their collective decision-making process through voting (classification) or averaging (regression). Random Forest is renowned for its high accuracy, adaptability to various dataset types, and capacity for automatic feature selection. It is extensively utilized in a variety of industries, including finance, healthcare, and image analysis, for tasks like feature importance ranking, regression, and classification.

4.3. Naïve Bayes

Based on Bayes' theorem, the Naive Bayes machine learning model determines the likelihood of a hypothesis

given the available data. It is especially well-liked for tasks involving classification. To simplify computations, Naive Bayes makes the "naive" assumption that features are conditionally independent given the class label.

The Bayes theorem formula is:

$$P(\text{Class}|\text{Features}) = \frac{P(\text{Features}|\text{Class}) \cdot P(\text{Class})}{P(\text{Features})}$$

Here,

$P(\text{Class}|\text{Features})$ is the class's posterior probability given its features.

$P(\text{Features}|\text{Class})$ is the probability of noticing the characteristics given the class.

$P(\text{Class})$ is the class's prior probability.

$P(\text{Features})$ is the likelihood of seeing the characteristics.

The likelihood term for the naive assumption is factorized as the sum of the probabilities of each individual feature:

$$P(\text{Features}|\text{Class}) = P(\text{Feature1}|\text{Class}) \cdot P(\text{Feature2}|\text{Class}) \cdot \dots$$

Particularly for text classification and spam filtering, Naive Bayes is a computationally efficient method that frequently yields good results. Its performance in complicated, correlated feature scenarios may be impacted if the independence assumption is not met in every situation.

5. ANALYSIS AND RESULTS

	Users count	Activity types count	Workouts count	Total records count
0	1,104	49	253,020	111,541,956

Fig 5.1 Overall dataset summary

The fitness dataset's synopsis, which includes information on user demographics, activity categories, and workout statistics, is shown above.

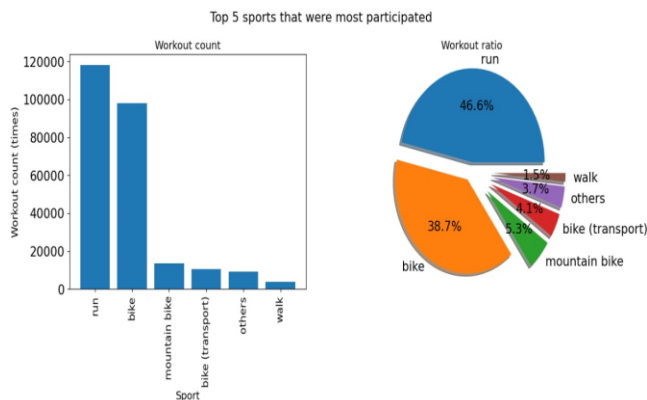


Fig 5.2 Top 5 sports that were most participated

The proportions of the top 5 varietales in the dataset were computed for this study. Two methods were used to visualize these proportions: a bar plot that ranked the percentages on the y-axis based on popularity and the exercise type on the x-axis, and a pie chart that displayed the top five categories scaled by their corresponding percentages. These visualizations reduce complicated data into simply understood images by giving a brief and concise summary of the most common exercise kinds in the dataset.

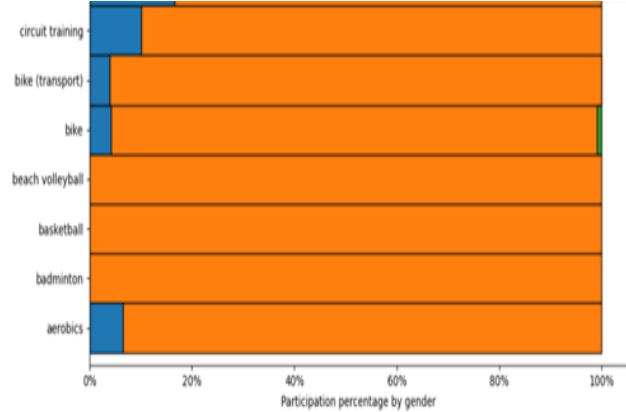


Fig 5.3 Sport wise participation percentage by gender

Here we calculate and visualize activity participation by gender. It counts the activities and creates a stacked horizontal bar chart to display the participation percentage of each gender in different sports from the given DataFrame. The code also customizes the chart's appearance and formatting for clarity.

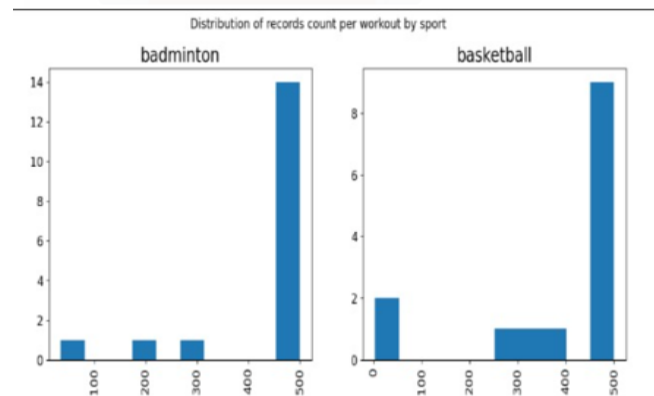


Fig 5.4 Distribution of records count per by sports.

The distribution of the number of recordings per workout for different sports is shown by the histograms. It is clear that different sports have different records for each session. Workout records for most sports are centred around a count of 500, but others show differing distributions, suggesting

variations in data accessibility, and recording techniques for various activities.

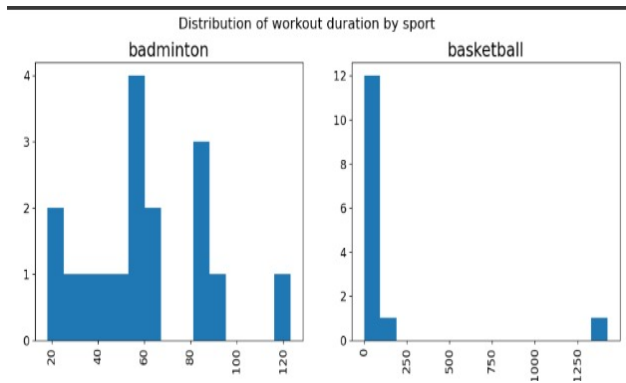


Fig 5.5 Distribution of workout duration by sport

These histograms by highlighting the most typical workout durations for each sport, provide an overview of activity patterns and aid in identifying outliers or abnormalities in the data

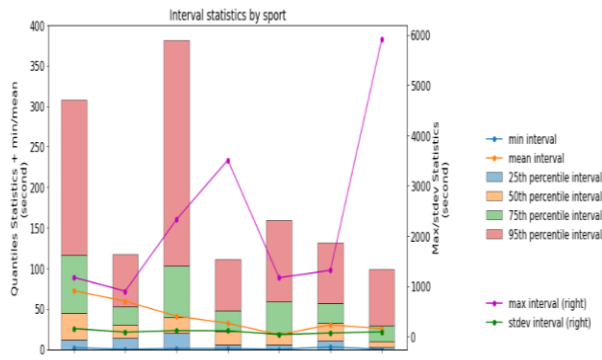


Fig 5.6 Interval Statistics by sports

For quantile statistics, this visual representation uses bar plots, and for minimum, mean, maximum, and standard deviation statistics, it uses line graphs. An extensive examination of interval data is made possible by the figures, which offer a thorough grasp of how these statistics differ amongst sports.

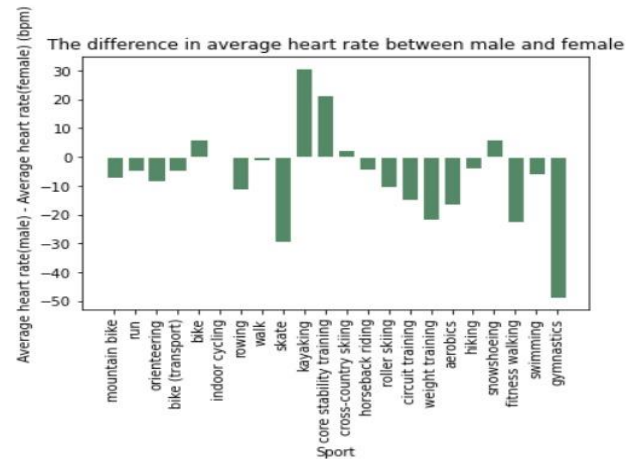


Fig 5.7 The difference in average heart rate between male & Female

We have taken data out of a DataFrame, transformed it into an RDD (Resilient Distributed Dataset), and shown the differences in average heart rate between the various sports. The disparities in the average heart rates of men and women for various sports were then clearly compared using a bar chart that we made.

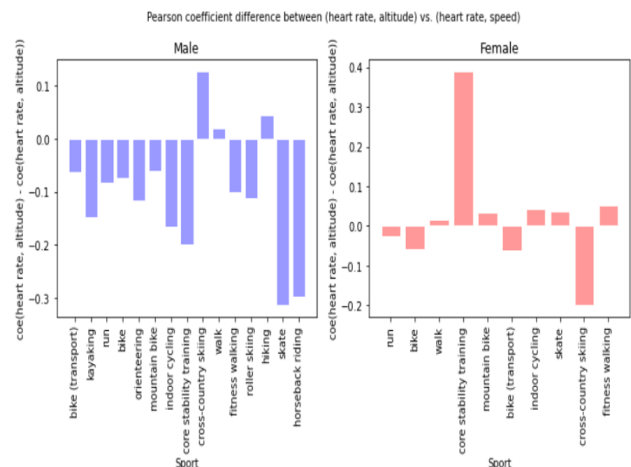


Fig 5.8 Pearson coefficient difference between (heart rate, altitude) vs. (heart rate, speed)

We evaluated the impact of speed and altitude changes on heart rate patterns by examining these correlations. This analysis allowed us to compare the relationships between genders and provided insights into the physiological response to various forms of physical activity.

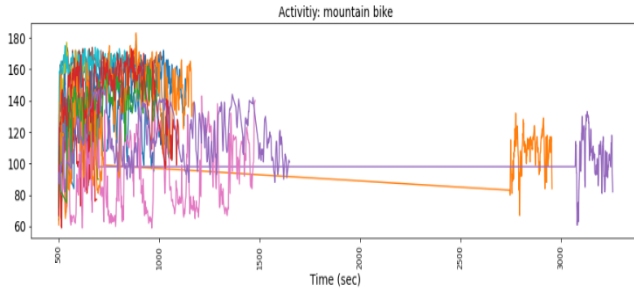


Fig 5.9 Heart rate pattern associated with various sports

By determining the length of each timestamp record from the first record of a workout, we were able to normalize the time for all workouts. After classifying the data by sport, it plots the heart rate on this normalized time. The resulting graphic is made up of several subplots that each display the heart rate patterns associated with various sports. For easier understanding, the chart has been customized with labels, legends, and titles.

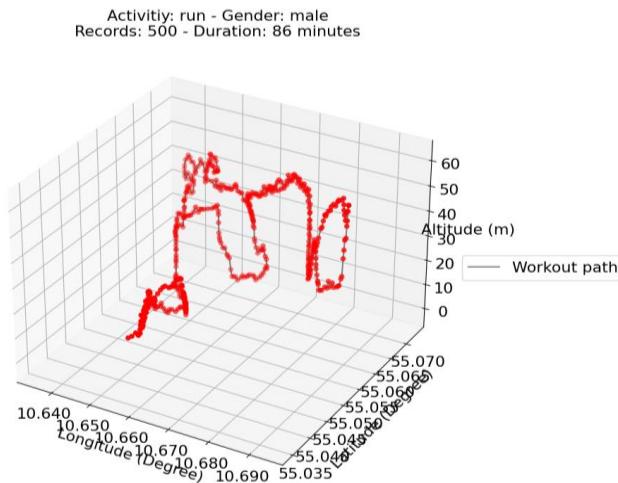


Fig 5.10 workout path in 3D graphs for each sport

We created 3D visualizations of users' workout routes using geographical data such as latitude, longitude, and altitude. These 3D workout route plots can provide a visual representation of how users move in geographic space during their workouts, which can be useful for understanding patterns in the data.

The dataset was divided into two parts: a training set with 70% of the data and a testing set with the remaining 30%. The training set was used to train the machine learning model, and the testing set was used to assess the model's performance on new and previously unseen data. This separation aided in determining the model's ability to generalize to previously unseen examples.

We used important input features like gender, geographic coordinates (latitude, longitude, altitude), heart rate, and derived speed in our heart anomaly prediction model. The highest heart rate recorded during each workout is used to determine the binary target variable (0 or 1). By using a combination of physiological and location-related indicators, this approach enables us to predict whether heart anomalies are present (1) or absent (0).

We used the PySpark library to run Logistic Regression on the provided data to train a model. The trained model was then used to make predictions on the test data, and its performance was evaluated using accuracy. Overall, 41% of accuracy is achieved with it.

We trained a Random Forest classifier on a labelled training dataset, which then predicts on a separate test dataset and assesses the model's accuracy using the specified metric. For 95% of the instances in the test dataset, this model correctly predicted the target variable.

We trained a Gaussian Naive Bayes classifier on a training dataset, used the trained model to predict on a test dataset, and evaluated the model's accuracy using multiclass classification accuracy. On the test dataset, the Naive Bayes model achieved an accuracy of 85%.

6. COMPARISON OF ML MODELS

Random Forest's ensemble nature and capacity to identify intricate relationships in the data may allow it to predict heart anomalies better than Logistic Regression and Naive Bayes. Below are the accuracy scores for each of the three machine learning models.

Table 6.1. ML model comparisons

Model	Test Accuracy
Logistic regression	0.417
Random Forest	0.956
Naïve Bayes	0.857

Of the three machine learning models, the Random Forest algorithm achieved the highest accuracy, outperforming both Naive Bayes and Logistic Regression. Reducing overfitting and improving generalization, Random Forest blends several decision trees, each trained on a distinct subset of the data and features. Intricate patterns and interactions in the dataset can be modelled by Random Forest thanks to this ensemble approach, which also helps to mitigate the drawbacks of simpler models like Logistic Regression and Naive Bayes. Random Forest is also an excellent method for handling non-linearity and feature interactions, which is important for medical datasets where relationships can be complicated. When compared to single models like Logistic Regression and Naive Bayes, Random Forest's superior accuracy in predicting heart anomalies is

often attributed to its inherent diversity and robustness to outliers and missing values.

7. CONCLUSION:

In conclusion, this study examined the methods used by the fitness app Endomondo for gathering and analyzing data, with a particular emphasis on heart rate anomaly detection. We saw important insights into fitness trends and their relationship to health outcomes from the thorough data visualizations and analyses. The top five sports were determined, distributions of workout records and durations were examined, and gender-based participation percentages were represented graphically. While heart rate patterns related to different sports and their 3D representations offered insightful information, interval statistics gave a thorough understanding of workout trends. We used machine learning algorithms, such as Random Forest, Naïve Bayes, and Logistic Regression, to find irregular heart rate patterns that might be signs of cardiac abnormalities. Our investigation provided new information on activity preferences, usage patterns, and relationships between fitness and health outcomes. Compared to Logistic Regression (41%) and Naïve Bayes (85%), the Random Forest model showed higher accuracy (95%). Random Forest's ensemble nature, which captures complex relationships in data, worked well for heart anomaly prediction. This work creates a methodological basis for preventive health screening using commercial fitness data and gives developers a foundation to improve fitness app features.

ACKNOWLEDGMENTS

We like to thank all the authors for their contributions to this study. Tanuj managed the introduction, data analysis, and report coherence; Dharani handled methodology and statistical analysis; Hrushika specialized in data visualization and project coordination; and Shivani focused on data collection and organization. Together, we produced a comprehensive report.

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